

C13-C14

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Instruire asistată de calculator C13-C14

Serious games in education/ Jocuri de formare in educatie

- Generalități, exemplificări.
- Avantaje/dezavantaje.
- Discutii.

EXERCİȚIU

Intra pe Menti.com si introdu **3173 5014**.

“ Inainte sa vorbim despre jocurile serioase, ce jocuri pe calculator cunoașteți? ”



Ce vom face?



01 Descoperim

Ce sunt jocurile serioase, de formare sau serious games?

✓ Exercițiu colaborativ in Jamboard!

02 Invatam

Incearca si tu.

03 Fixam

Fixare prin discuții in grupuri de lucru (breakout rooms/face-to-face)

04 Recapitulam la final

Generalități, exemplificări

Ce sunt jocurile de formare (engl. Serious games)

Sunt aplicații de tip joc pe calculator/dispozitiv mobil/browser/televizor ale căror scop principal declarat este de a forma deprinderi reale, utilizate în viața de zi cu zi sau în viața profesională.

Observatie

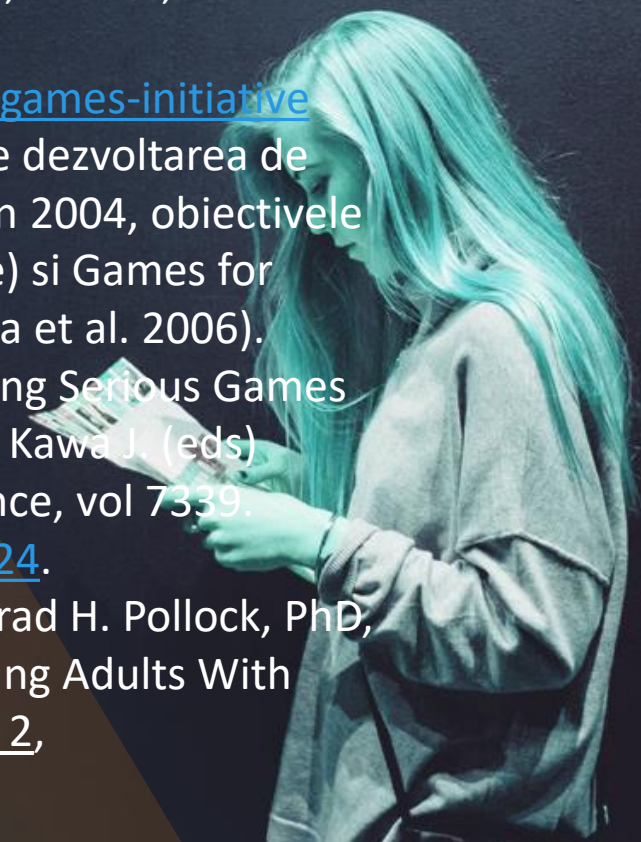
“Gamification” este procesul de a adauga functionalitati similare jocurilor pe calculator in alte tipuri de aplicatii/contexte (site-uri web, aplicatii specifice anumitor afaceri) cu scopul de a favoriza și incuraja aderarea, implicarea și colaborarea utilizatorilor.

DuoLingo



Resurse bibliografice

1. Alessandro De Gloria, Francesco Bellotti, Riccardo Berta, *Serious Games for education and training*, published in 2014 in International Journal of Serious Games, vol 1, Issue 1, ISSN: 2384-8766, DOI:<https://doi.org/10.17083/ijsg.v1i1.11>
2. Serious Games Initiative - <https://www.wilsoncenter.org/program/serious-games-initiative>
In 2002, a fost lansat proiectul "Serious Games Initiative" pentru a sustine dezvoltarea de jocuri care sa adreseze aspecte de management si politici de business. Din 2004, obiectivele au fost extinse spre Games for Change (aspect sociale si schimbari sociale) si Games for Health pentru a incuraja dezvoltarea de aplicatii pentru medicina (Annetta et al. 2006).
3. Szczesna A., Tomaszek M., Wieteska A. (2012) The Methodology of Designing Serious Games for Children and Adolescents Focused on Psychological Goals. In: Piętka E., Kawa J. (eds) Information Technologies in Biomedicine. Lecture Notes in Computer Science, vol 7339. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-31196-3_24.
4. Pamela M. Kato, PhD, EdM; Steve W. Cole, PhD; Andrew S. Bradlyn, PhD; Brad H. Pollock, PhD, MPHA Video Game Improves Behavioral Outcomes in Adolescents and Young Adults With Cancer: A Randomized Trial, publicat in Pediatrics, 2008 Volume 122, Issue 2, <https://doi.org/10.1542/peds.2007-3134>



Exemple – jocuri de formare pentru copii

❖ Pick Your Plate!

Creat de către și pentru Smithsonian Science Education Center/ Segment de varsta: clasele 4-8

URL: <https://ssec.si.edu/pick-your-plate>

Pick Your Plate! A Global Guide to Nutrition is an educational nutrition game that will help teach students about building healthy meals while using nutritional guidelines from countries around the world!

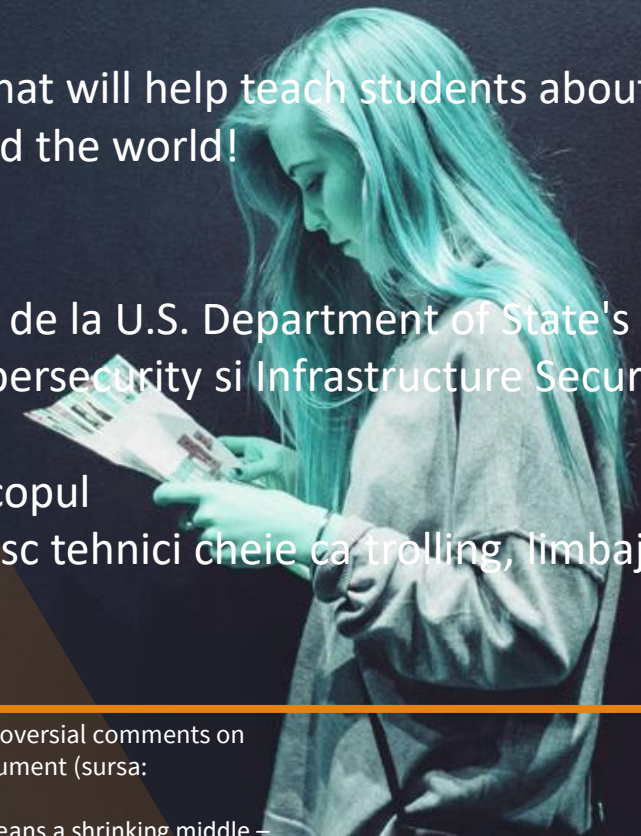
❖ Breaking Harmony Square

Creat de University of Cambridge și DROG (Dutch media collective) cu fonduri de la U.S. Department of State's Global Engagement Office (GEC) , U.S. Department of Homeland Security's Cybersecurity și Infrastructure Security Agency (CISA)/ Segment de varsta: 15+.

Personajul principal este ofiterul șef pentru dezinformare din Piața Armoniei. Scopul jocului este de a înțelege cum anume se răspândește dezinformarea. Se folosesc tehnici cheie ca trolling, limbaj emotional și polarizarea audienței**.

*Trolling – (verb), as it relates to internet, is the deliberate act, (by a Troll – noun or adjective), of making random unsolicited and/or controversial comments on various internet forums with the intent to provoke an emotional **knee jerk** reaction from unsuspecting readers to engage in a fight or argument (sursa: www.urbandictionary.com)

** Polarization means more people are aligning themselves with one side of a divide, and positioning themselves against the other. It also means a shrinking middle – fewer people who split their ticket at the polls or are willing to listen to arguments with which they disagree.(sursa: <https://www.americanpressinstitute.org/publications/reports/strategy-studies/contending-with-polarized-audiences/>)



EXERCİȚIU

Individual, 3-4 minute



Cauta alte exemple de jocuri de formare pentru copii.

Postează rezultatele: numele tau, nume joc si link in Class Notebook, zona colaborativa.



Alte jocuri de formare pentru copii

- ❖ <https://codecombat.com/>
- ❖ <https://variant.triseum.com/> - o alta forma de a invata Analiza matematica
- ❖ (sursa: <https://www.seriousgamemarket.com/2017/05/new-serious-game-transforms-calculus.html>)
- ❖ <https://www.memo-juegos.com/>
- ❖ <https://www.memo-juegos.com/juegos-simon/juego-de-simon-setas>



Exemple – jocuri de formare pentru adulți

❖ SQL Murder Mystery

URL: <https://mystery.knightlab.com/>

"A crime has taken place and the detective needs your help. You vaguely remember that the crime was a murder that occurred sometime on Jan.15, 2018 and that it took place in SQL City. Start by retrieving the corresponding crime scene report from the police department's database."

Utilizatorul trebuie sa foloseasca notiunile si competentele SQL pentru a gasi ucigasul.

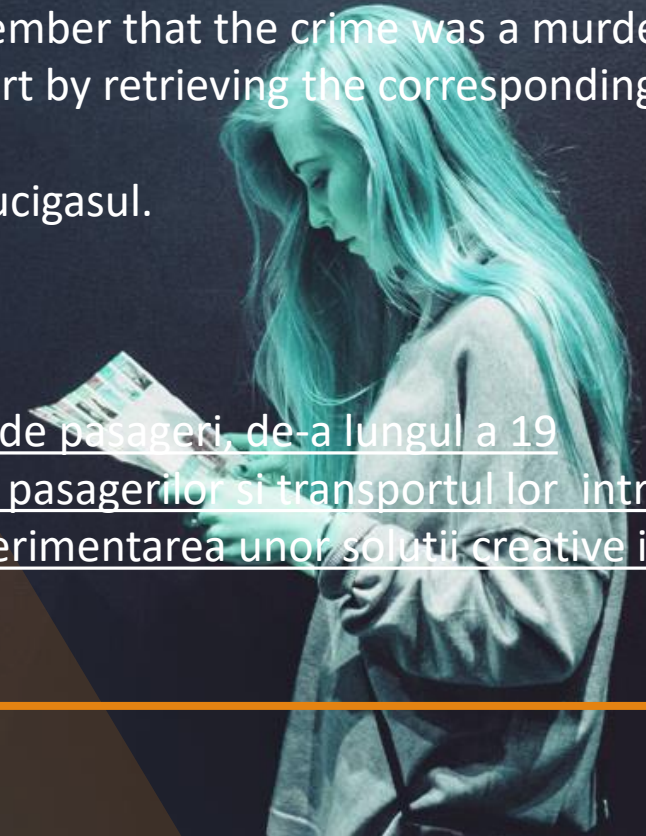
❖ Elevator Saga

URL: <https://play.elevatorsaga.com/>

Utilizatorul programeaza o serie de lifturi pentru gestiona un numar crescand de pasageri, de-a lungul a 19 incercari. Algoritmul trebuie adaptat pentru a minimiza timpul de asteptare al pasagerilor si transportul lor intre etaje. Jocul exerseaza puternic script JavaScript si forteaza identificarea si experimentarea unor solutii creative in procesul de optimizare.

https://youtu.be/J1ie70_rDXE

***sursa:<https://www.freecodecamp.org/news/best-coding-games-online-adults-learn-to-code/>



EXERCİȚIU

Individual, 3-4 minute



Cauta exemple de jocuri de formare pentru adulti.
Postează rezultatele: numele tau, nume joc si link
in Class Notebook, zona colaborativa.



Exemple – jocuri de formare pentru adulți

❖ EDUardo

URL: <https://edu-simulation.com/>

<https://www.youtube.com/watch?v=tTtHRuDeC6g>

"Enhance curriculums using EDUardo

EDUardo offers students the opportunity to practice in an interactive business simulation".

"Simulation participants can learn how to analyze the basic elements of business operation by managing their own virtual company in a risk-free environment."

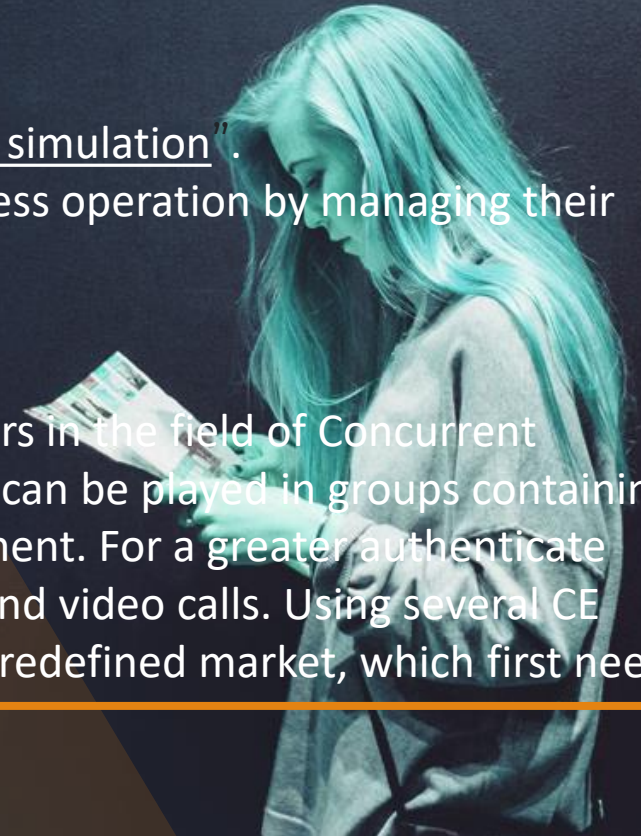
❖ COSIGA

URL: <https://www.biba-gaminglab.com/en/our-games/cosiga/>

"COSIGA is a multiplayer simulation game for students, engineers and managers in the field of Concurrent Engineering (CE). It was developed in the European project "COSIGA". COSIGA can be played in groups containing up to 5 players, who will be asked to cooperate as a team in product development. For a greater authentic players can be placed in separate locations and communicate via chat, email and video calls. Using several CE principals the objective of the game is to produce a cost-effective truck for a predefined market, which first needs to be analysed by the players."

❖ Observatie

<https://www.alliedmarketresearch.com/serious-games-market>



Exemple – jocuri de formare pentru adulți

❖ RefQuest

URL: <https://refquest.westernsydney.edu.au/play.html>

The main objective of refQuest is the support of idea-generation in a structured way. The game shall stimulate this goal by looking at a given problem from many perspectives. The perspective of the game changes in each round (Baalsrud Hauge et al. 2007).

❖ <https://www.lumosity.com/>

❖ Observatie

<https://www.alliedmarketresearch.com/serious-games-market>



EXERCİȚIU

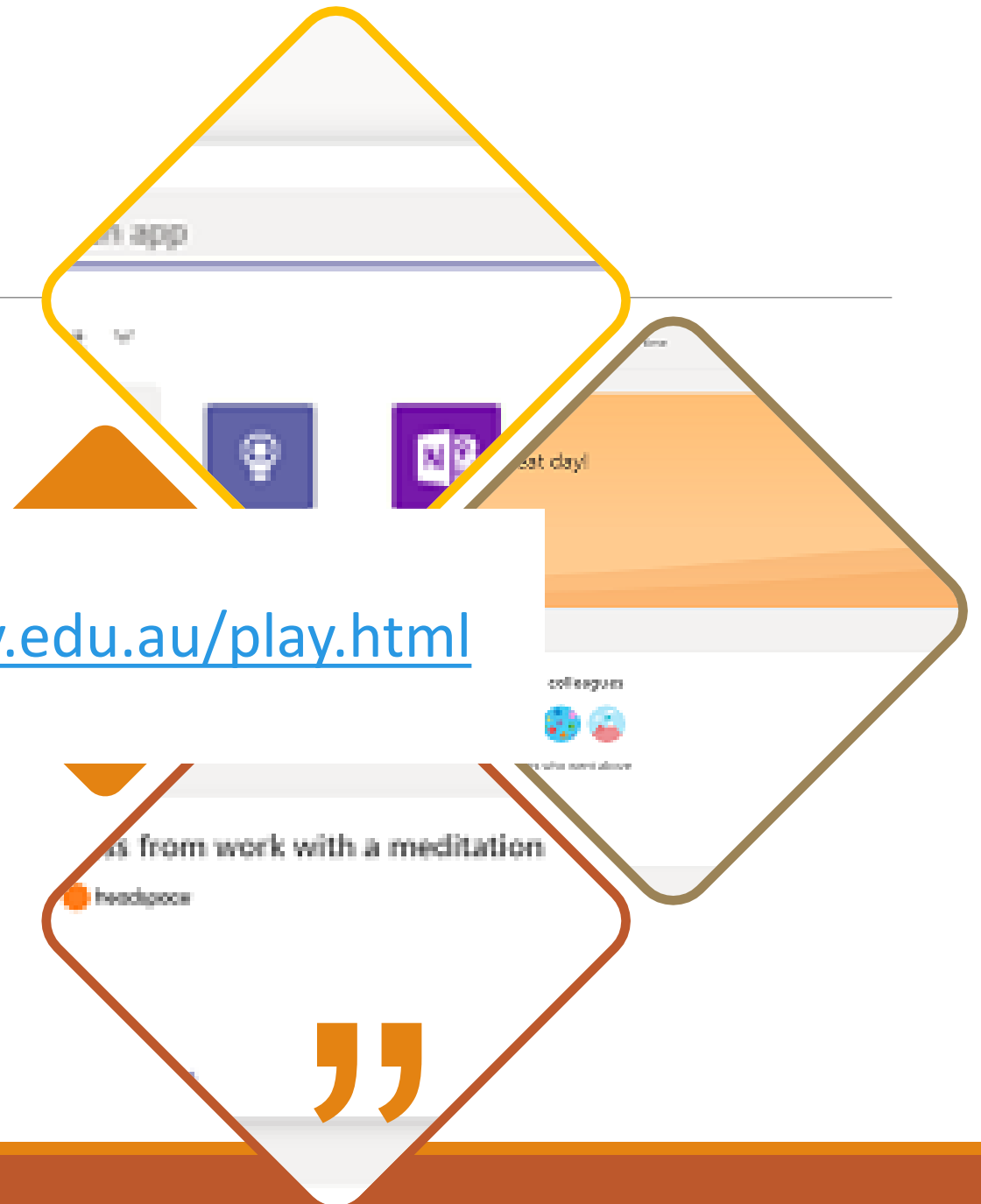
7-10 minute, cu dispunere in Breakout rooms



Acceseaza site-ul .

<https://refquest.westernsydney.edu.au/play.html>

Incearca jocul. Ai 7 minute.



Discutie deschisa



Discută observațiile tale asupra jocului.



Avantaje

- Recrearea unor context realiste care sa incurajeze invatarea cu costuri mai reduse decat mediul real
- Reducerea riscurilor de accidentare
- Cresterea sigurantei utilizatorului in perioada de invatare/training
- Incurajeaza colaborarea online, utilizarea de comenzi verbale, multi-tasking-ul
- Sustin abordarea, explorarea si solutionarea problemelor.



Dezavantaje

- Dezavantajeaza utilizatorii care nu au deprinderi de jocuri pe calculator
- Pot necesita echipament dedicat costisitor.
- Poate necesita capabilitati digitale.
- Poate duce la extreme in timpul alocat jocului/dependenta.
- Lipsa contactului uman nemijlocit.





Recap

Ce instrumente am folosit azi? Interactiune pe jamboard.

01

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02

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03

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04

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05

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Mulțumesc pentru atenție!

Ne auzim la examen!

